Battery Management System – High-Level Design

AUTOSAR Classic Platform | Sample Reference Architecture Document

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Prepared for internal engineering review and pilot evaluation

AUTOSAR Classic Platform | High-Level Design Document

Document Control

Field	Details
Document Title	Battery Management System – High-Level Design
Version	1.1
Document Type	Software Architecture / High-Level Design (HLD) Document
Purpose	Describe the software architecture, cell-monitoring interfaces, signal flow, and dependencies of the Battery Management System for an EV traction battery pack.
Intended Audience	System architects, functional safety engineers, software developers, test engineers
Status	Sample / Reference (Synthetic Data)
Confidentiality	Internal use (sample/synthetic data for pilot evaluation)

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Introduction and Scope

This document describes the High-Level Design of the Battery Management System (BMS) software for an electric vehicle traction battery pack. The BMS monitors cell voltage, temperature, and state-of-charge, and manages contactor control and thermal balancing. This revision (1.1) adds the Cell Balancing Controller component and updates the State-of-Health estimation interface.

Scope covers the following software components:

- SWC_CellVoltageMonitor
- SWC_ThermalMonitor
- SWC_StateOfChargeEstimator
- SWC_ContactorController
- SWC_CellBalancingController (new in v1.1)

Software Component Overview

Component Name	Responsibility	Mapped ECU
SWC_CellVoltageMonitor	Reads and validates individual cell voltages via daisy-chain ADC	BMS_ECU_01
SWC_ThermalMonitor	Monitors pack thermistor network, detects over-temperature	BMS_ECU_01
SWC_StateOfChargeEstimator	Estimates SoC/SoH using coulomb counting and OCV lookup	BMS_ECU_01
SWC_ContactorController	Controls main and pre-charge contactors	BMS_ECU_01
SWC_CellBalancingController	Passive balancing control across cell groups (new)	BMS_ECU_01

Ports and Interfaces

Port Name	Direction	Interface Type	Owning Component
RPort_CellVoltageRaw	Required	Sender-Receiver	SWC_CellVoltageMonitor
PPort_CellVoltageValidated	Provided	Sender-Receiver	SWC_CellVoltageMonitor
RPort_ThermistorRaw	Required	Sender-Receiver	SWC_ThermalMonitor
PPort_PackTemperature	Provided	Sender-Receiver	SWC_ThermalMonitor
RPort_CellVoltageValidated	Required	Sender-Receiver	SWC_StateOfChargeEstimator
PPort_SoCEstimate	Provided	Sender-Receiver	SWC_StateOfChargeEstimator
PPort_SoHEstimate	Provided	Sender-Receiver	SWC_StateOfChargeEstimator (new in v1.1)
CSPort_ContactorDriverService	Required	Client-Server	SWC_ContactorController
RPort_CellVoltageValidated	Required	Sender-Receiver	SWC_CellBalancingController (new)

Port Name	Direction	Interface Type	Owning Component
PPort_BalancingCmd	Provided	Sender-Receiver	SWC_CellBalancingController (new)

Signals and Data Elements

Signal Name	Data Type	Range / Values	Description
CellVoltageValidated	uint16[96]	2500–4300 mV per cell	Validated per-cell voltage array
PackTemperature	int16[12]	-40 to 85 degC	Per-zone thermistor readings
SoCEstimate	uint8	0–100 %	State of charge estimate
SoHEstimate	uint8	0–100 %	State of health estimate (new in v1.1)
ContactorCmd	uint8	0 = Open, 1 = Pre-charge, 2 = Closed	Contactor control command
BalancingCmd	boolean[96]	TRUE / FALSE per cell	Passive balancing enable per cell (new)

Component Dependencies

Component	Depends On	Dependency Type
SWC_CellVoltageMonitor	BSW_SPI (daisy-chain AFE)	Raw ADC data acquisition
SWC_ThermalMonitor	BSW_ADC	Thermistor analog input
SWC_StateOfChargeEstimator	SWC_CellVoltageMonitor, BSW_NvM	Voltage input, persisted coulomb counter
SWC_ContactorController	SWC_ThermalMonitor, SWC_StateOfChargeEstimator	Safety interlock conditions
SWC_CellBalancingController	SWC_CellVoltageMonitor	Per-cell voltage deviation input (new)

Functional Flow – Contactor Closing Sequence

- 1. On ignition-on request, SWC_ContactorController checks PackTemperature within safe range.
- 2. SoCEstimate and SoHEstimate are checked against minimum operating thresholds.
- 3. Pre-charge contactor is closed first via CSPort_ContactorDriverService.
- 4. Bus voltage is monitored to confirm pre-charge completion before main contactor closure.
- 5. Main contactor is closed; ContactorCmd transitions to Closed state.
- 6. SWC_CellBalancingController begins evaluating per-cell voltage deviation for balancing (v1.1 addition).

Change Summary (v1.0 to v1.1)

- Added SWC_CellBalancingController component and its ports/signals.
- Added PPort_SoHEstimate to SWC_StateOfChargeEstimator (previously SoC-only in v1.0).
- No changes to SWC_ContactorController interfaces in this revision.

Assumptions and Open Points

- Balancing current limit calibration is placeholder data pending thermal validation.
- SoH estimation algorithm accuracy has not been validated against aged cell data.
- Assumes a 96-cell series pack configuration; other topologies are out of scope.

Revision History

Version	Date	Description
1.0	July 2026	Initial release – voltage, thermal, SoC, contactor components
1.1	September 2026	Added Cell Balancing Controller and SoH estimate interface